HUDSON
Products Corporation

Tuf-Lite® and Tuf-Lite II® Fans 6000 Series Hub

INSTALLATION MANUAL



Adjustable Pitch Fan Assembly
32’ and 33’ Diameter

Hudson Tuf-Lite® and Tuf-Lite II® fan blades

Hudson Tuf-Lite® (black) fan blades are made from fiberglass reinforced epoxy resin having a very high strength-to-weight ratio and corrosion resistance (not shown).

Hudson Tuf-Lite II® (white, prev. blue***) fan blades are made from fiberglass reinforced vinyl-ester resin having a very high strength-to-weight ratio and superior ultra-violet and corrosion resistance. An elastomeric blade/holder joint cover (not shown) prevents moisture from entering the blade (shown above).

The individually balanced blades can be replaced independently - matched sets are not required.

RECOMMENDED TOOLS

- Long T-Handle Allen Wrench Set (3/16" to 3/8")
- Medium Size Flat Head Screw Driver
- Brass Ball Peen Hammer
- Flat Bastard File
- 240 Grit Sand Paper
- Anti-Seize Lubricant
- WD-40
- 12" Crescent Wrench
- Shop Towels
- Exact-A-Pitch® Digital Protractor (P/N 62375)
- 25 ft. Measuring Tape
- Pencil or Marker
- Open/Box End Wrench Set (1/2" - 1-1/2")
- Socket Set for 1/2" Drive (1/2" - 1-1/2")
- Torque Wrench(s) Rated for 0-200 ft-lb.

INSTALLATION PROCEDURES

ASSEMBLY WITH BUSHING

Clean all mating surfaces between hub, bushing and shaft. All grease and lubricant should be removed, leaving the mating surfaces dry.

If there is no shoulder on shaft to prevent bushing from sliding down shaft, slide spacer/sleeve (not provided) on shaft before bushing or use a thrust retainer (optional equipment) on top of hub. Slide bushing and key onto shaft until flush with end of shaft. The shaft size determines the bushing type (U1). Lock bushing on shaft by tightening the set screw in flange with an Allen Wrench. Line up key and set hub on bushing. Engage the three (3) cap screws in flange of bushing into hub spool, using a torque wrench with a socket, and tighten evenly. Use the following table to determine the proper tools and torque values.

Bushing Size	Allen Wrench Size	Cap Screw Size	Socket Size	Torque (ft-lb) Dry
U1	3/16"	5/8"	15/16"	140

ASSEMBLY WITH STRAIGHT SHAFT (NO BUSHING)

Clean all mating surfaces between the hub and the shaft. If there is no shoulder on shaft to prevent hub from sliding down shaft, slide spacer/sleeve (not provided) on shaft before hub or use a thrust retainer (optional equipment) on top of hub. Install key in shaft. Line up key and keyway and set hub on shaft. Tighten set screw(s) in hub.

ASSEMBLY WITH TAPERED SHAFT (NO BUSHING REQUIRED)

Clean all mating surfaces between the hub and shaft.

Align keyways and install hub. Install retainer plate and cap screw(s) with lock washer(s). Shaft size determines what size cap screw is necessary. Using a torque wrench with a socket, evenly tighten cap screw to recommended standard per table below.

Cap Screw Size	Socket Size	Torque Value (ft-lb)	
		Lubricated	Dry
5/8" NC	15/16"	100	110
3/4" NC	1-1/8"	125	130
1" NC	1-1/2"	150	160

NOTE: Retaining arrangement varies with gear shaft design.

THRUST RETAINER (optional equipment)

Install proper load bolt (not provided) into top of fan shaft and tighten (See Figure 1). Install thrust retainer channel on top hub plate using existing hub spool cap screws. Torque cap screws to 60-65 ft-lb. Install thrust retainer eyebolt and jam nut. Hand tighten eyebolt. Tighten jam nut securely against top of thrust retainer channel.

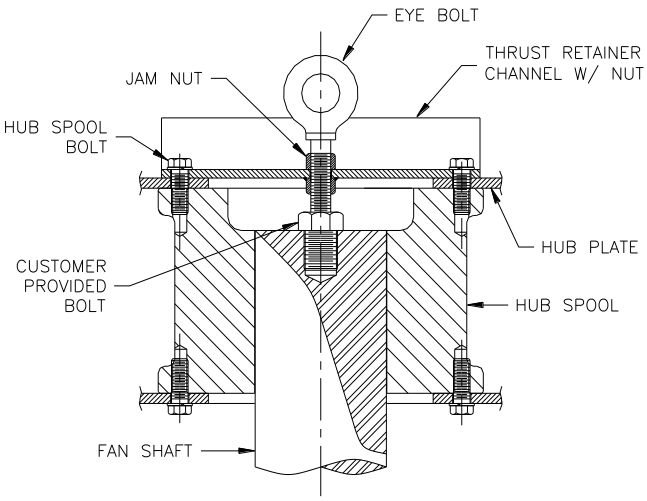


Figure 1

BLADE INSTALLATION

Remove blade clamp bolts, nuts, lock washers, and blade clamp halves from hub. Assemble blade clamp halves over groove in blade neck, and install into hub (See Figure 2). The thick leading edge will be to your left and thin trailing edge will be to your right as you stand at end of blade.



Figure 2

Install clamp bolts through hub plates and blade clamp, putting bolt heads on top, lock washers and nuts on bottom. Tighten lightly (See Figure 3).



Figure 3

SET PITCH AND TRACK

Use Hudson EXACT-A-PITCH® digital protractor (See Figure 4) or a bubble protractor to set blade pitch. Mount protractor on a flat bar as a base and place it approximately 1" from tip of blade. Note pitch on protractor. Rotate fan 360°, noting high and low pitch readings. Locate place where pitch reading is at mid-point between high and low readings, and set pitch at that point.



Figure 4

Rotate blade in clamp until digital protractor shows specified pitch angle to within +/-0.2°. *Fan pitch angle is shown on fan specification sheet for design duty.* After desired pitch angle is set, raise and lower end of fan blade and find midpoint of blade travel. Hold blade at the midpoint. Pull blade outward so that the blade neck flange rests against the back of the blade clamps. Push blade to the right to remove all slack.

Use torque wrench to tighten clamp bolts to 100 ft-lb (lubricated) or 125 ft-lb (dry). Re-check pitch setting. Blade must be set within +/-0.2° of desired pitch angle. Tighten clamp bolts evenly. ***DO NOT OVER-TORQUE CLAMP BOLTS.***

When bolts are tightened, hold a pencil against top end of blade and mark the level onto a fixed object, such as a pole or the fan ring.

Install remaining blades at same place as first blade, following the instructions above. After tightening bolts, mark top end of each blade in same place first blade was marked. If marks differ by 1" or more, adjust blade.

CHECK TRACK

After fan is installed in fan stack cylinder ring, outline top side of each blade onto fan stack cylinder ring with a marker (See Figure 5). The difference between levels of highest and lowest outlines should not be more than 1". Correct blade track by loosening clamp bolts and adjusting blade to match track of other blades. Re-tighten bolts and re-check track and pitch angle setting. Re-tighten blade clamp bolts to recommended standard of 100 ft-lb (lubricated) or 125 ft-lb (dry) torque.



Figure 5

SEAL DISC ASSEMBLY & INSTALLATION

Install self adhesive rubber gaskets on both flanges of one seal disc half. Bolt two halves of seal disc together, using 3/8" NC bolts, flat washers, lock washer, and nut. Torque to 15 ft-lb (lubricated) and 20 ft-lb (dry).

Install 3/8" NC bolts at six (6) places on top hub plate (See Figure 6). Threaded portion of bolts must be pointing up to mount seal disc. Install lock washer, nut, and flat washer on each bolt. Tighten 3/8" NC nuts to 15 ft-lb (lubricated) and 20 ft-lb (dry).

Locate the six (6) mounting holes in seal disc and install over the six (6) bolts pointing up on upper hub plate. If difficulty is encountered, loosen bolts on seal flanges until seal disc can be mounted, then re-tighten to 15 ft-lb (lubricated) or 20 ft-lb (dry).

NOTE: The purpose of the seal disc is to prevent hot air from recirculating back down through the hub, increasing efficiency.

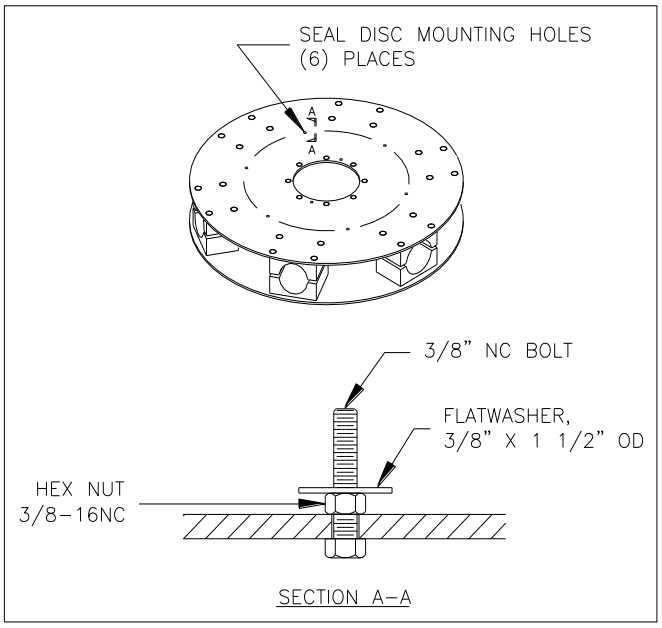


Figure 6

After mounting, install flat washer, lock washer, and 3/8" NC nuts. Tighten to 15 ft-lb (lubricated) or 20 ft-lb (dry). (See Figure 7)

Note: Refer to instructions included with seal disc for further details.

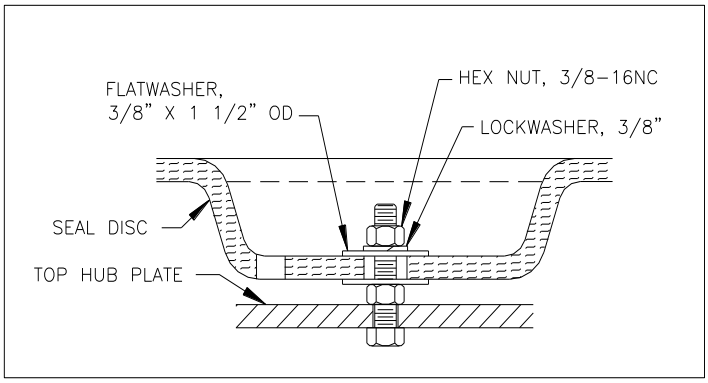


Figure 7

CHECKING TIP CLEARANCE

Rotate fan in position inside fan stack to check tip clearance (See Figure 8). The recommended tip clearance is between 1" and 1 1/2". Check for spots where fan blade clearance is not within the recommended tolerance. If necessary, adjust fan stack by shimming to obtain proper clearance.

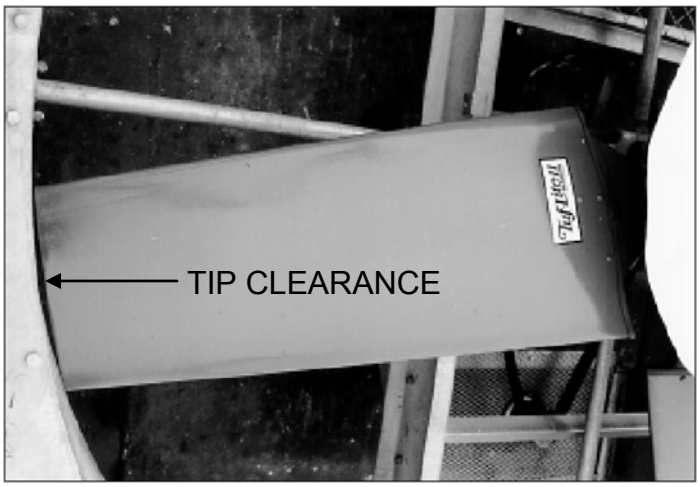


Figure 8

OPERATING INSTRUCTIONS

Start fan and check rotation. Viewed from top (discharge), fan blades should rotate clockwise.

Check motor power consumption to be sure fan is pulling desired load. *CAUTION: If positive pitch is set in summer to use all available motor amps (nameplate rating), motor could be overloaded in winter.* Design pitch angles usually do not use all of the available motor horsepower. This ensures that the motors will not be overloaded at low winter temperatures.

PART LIST
HUDSON PRODUCTS CORPORATION
Adjustable Pitch Fan Assembly 32' and 33' Diameter
Series 6000 HUB

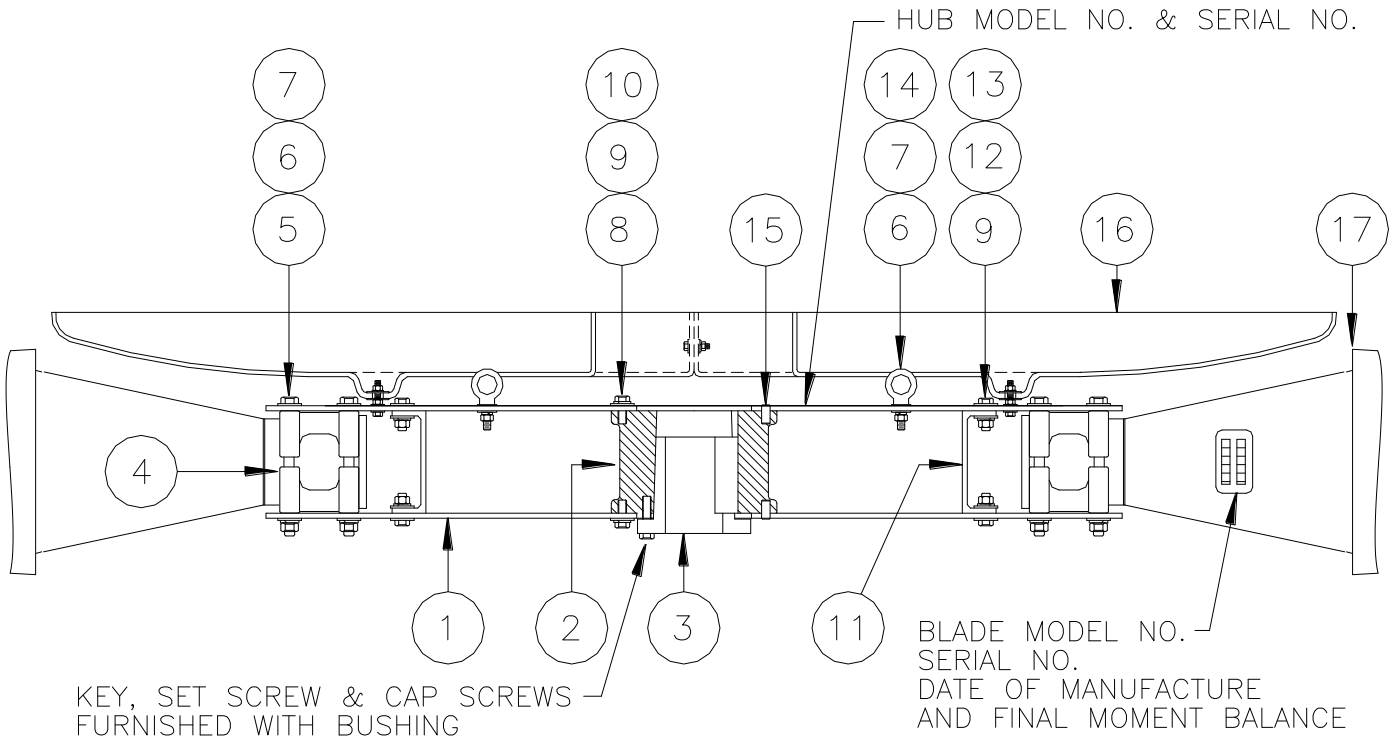
				NO. OF BLADES														
ITEM	DESCRIPTION	TYPE	PART. NO	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	32 Ft. Dia. Fan Shaft: 4.25" to 5.50" Dia.	U-1	Hub Assy. No. Part No.	6806 D6066	6807 D6067	6808 D6068	6809 D6069	6810 D6070	6811 D6071	6812 D6072	6813 D6073	6814 D6074	6815 D6075	6816 D6076	6817 D6077	6818 D6078	6819 D6159	6820 D6160
	33 Ft. Dia. Fan Shaft: 4.25" to 5.50" Dia.	U-1	Hub Assy. No. Part No.	6806S D6066S	6807S D6067S	6808S D6068S	6809S D6069S	6810S D6070S	6811S D6071S	6812S D6072S	6813S D6073S	6814S D6074S	6815S D6075S	6816S D6076S	6817S D6077S	6818S D6078S	6819S D6159S	6820S D6160S
1	72" Dia Hub Plate for 32' Fan (2 Per Hub)	U-1	Part No	D6046	D6047	D6048	D6049	D6050	D6051	D6052	D6053	D6054	D6055	D6056	D6057	D6058	D6059	D6060
	84" Dia Hub Plate for 33' Fan (2 Per Hub)	U-1	Part No	D6226	D6227	D6228	D6229	D6230	D6231	D6232	D6233	D6234	D6235	D6236	D6237	D6238	D6239	D6240
ITEM	DESCRIPTION		PART. NO	QUANTITY PER ASSEMBLY														
2	Hub Spool	U-1	65058	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
3	Bushing	U-1	Specify Bore	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
4	Blade Clamp Half, Powder Epoxy Coated Die Cast Alum (Standard) Option 1: Powder Epoxy Coated Ductile Iron** Option 2: Coal Tar Epoxy Coated Ductile Iron**		D5131 65013 65013C	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
5	Blade Clamp Bolt W/ Nut 3/4"-10 x 10" (Mech. Galv.)		79299	24	28	32	36	40	44	48	52	56	60	64	68	72	76	80
6	3/4" Flat washer (316 SS)		73720	54	62	70	78	86	94	102	110	118	126	134	142	150	158	166
7	3/4" Lock washer (Mech. Galv.)		73738	27	31	35	39	43	47	51	55	59	63	67	71	75	79	83
8	Hub Spool Cap Screw 5/8"-11 x 1 1/2" (316 SS)		72402	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
9	5/8" Flat washer (316 SS)		73719	48	48	48	48	48	48	48	48	48	48	48	48	48	48	48
10	5/8" Lock washer (316 SS)		73731	16	16	16	16	16	16	16	16	16	16	16	16	3	3	3
11	Stiffener Ring		D6004	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
12	Stiffener Ring Bolt w/ Nut 5/8"-11 x 1 3/4" (Mech. Galv.)		15347	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
13	5/8" Lock washer (Mech. Galv.)		73730	16	16	16	16	16	16	16	16	16	16	16	16	16	16	16
14	Eye Bolt W/ Nut 3/4"-10 x 2" (Mech. Galv.)		59547	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
15	Pin, Grooved, 1/2" X 1-1/2"		74540	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
16	112" Dia. Seal Disc Kit*		D6022	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
17	Tuf-Lite II® Blade (White)*** Tuf-Lite® Blade (Black)		(Varies) (Varies)	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

* Includes all hardware (316 SS) to assembly and mount.

** Recommended on concrete and round towers, or corrosive environments. Contact Hudson for pricing.

*** Blade color was blue prior to March 2006.

HUDSON PRODUCTS CORPORATION
Adjustable Pitch Fan Assembly 32' and 33' Diameter
Series 6000 HUB



STANDARD MATERIALS & FINISHES

Blades: Fiberglass reinforced vinyl ester or epoxy
Hub Spool: Ductile Iron, Zinc Rich Coating
Plates: Coal Tar Epoxy coated steel
Bushing: Malleable Iron
Seal Disc: Fiberglass Reinforced Polyester

Blade Clamps:
Powder Epoxy Coated Die Cast Alum (Standard)
Powder Epoxy Coated Ductile Iron (Option 1)
Coal Tar Epoxy Coated Ductile Iron (Option 2)
Fasteners:
Steel, Mech. Galvanized & 316 SS Opt.
Complete Fan with 316 SS (Option 1)
Complete Fan with K500 Monel (Option 2)

WHEN ORDERING, SPECIFY FAN DIAMETER, TYPE & NUMBER OF BLADES & SHAFT DIAMETER

EXAMPLE: APT — 32H — 8 — 4 1/2" BORE

APT	32H	8	4 1/2" BORE
Fan Model Adjustable Pitch	Fan Diameter & Blade Type (Specify "H" for Tuf-Lite II® Blades) (Uses 30H Blades) (Specify "B" for Tuf-Lite® Blades) (Uses 30B Blades)	Number of Blades	Shaft Diameter

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